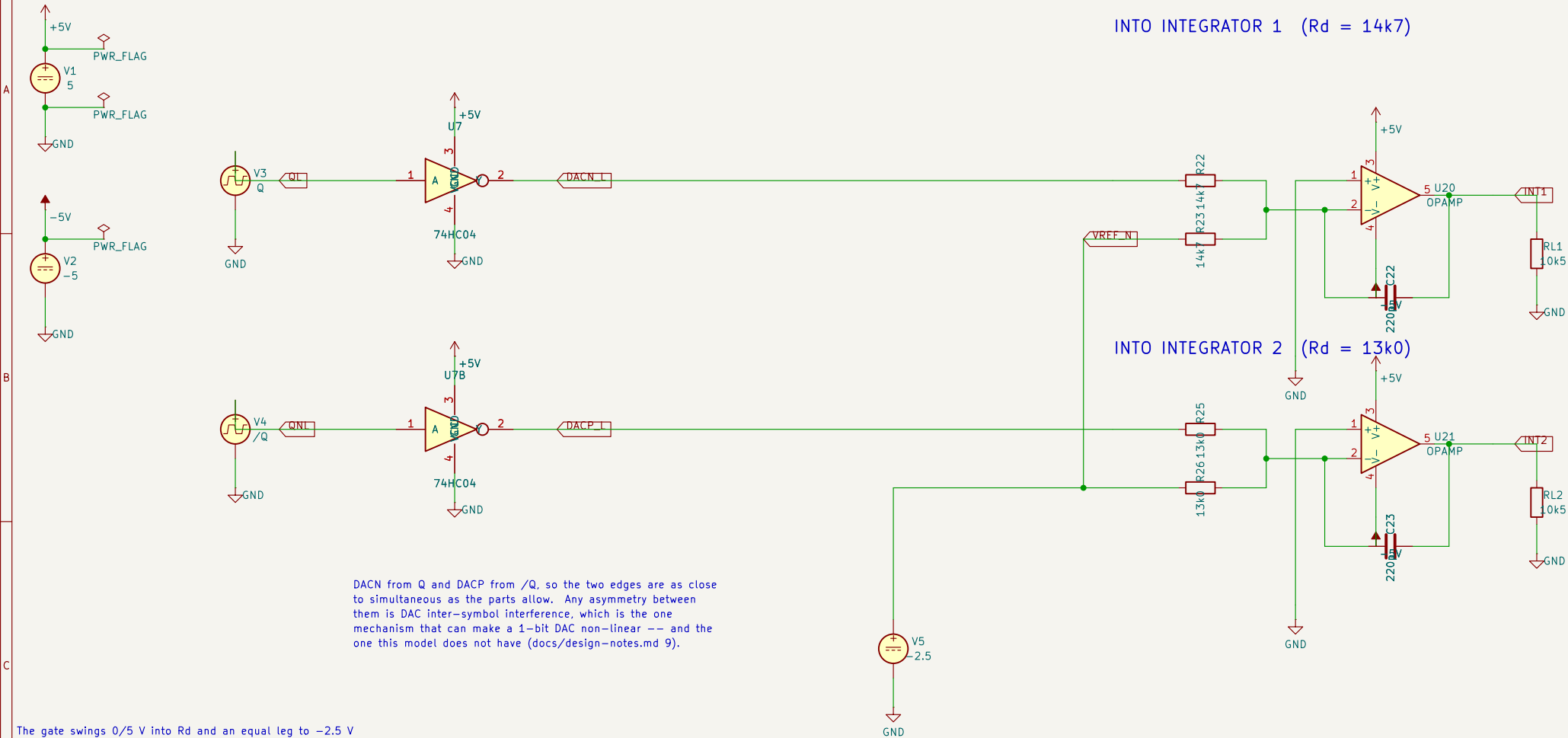


F The 1-bit DAC (74HC04 into 14k7 and 13k0, recentred by equal legs to -2.5 V)



DACN from Q and DACP from /Q, so the two edges are as close to simultaneous as the parts allow. Any asymmetry between them is DAC inter-symbol interference, which is the one mechanism that can make a 1-bit DAC non-linear -- and the one this model does not have (docs/design-notes.md 9).

The gate swings 0/5 V into Rd and an equal leg to -2.5 V recentres it, so the net feedback should be exactly +/-2.5 V -- which shows up as a triangle with equal up and down slopes of 2.5 / (Rd x 220p): 773 kV/s at integrator 1 and 874 kV/s at integrator 2.

Any inequality between the two slopes is the 74HC04's own 50 ohm of output resistance, which the ideal model in ../../sim does not have: it makes the high level 5 x 14700/14750 rather than 5 V, so the two DAC levels are not quite symmetric about the reference. That is a gain error and a DC offset, not distortion -- but it is worth knowing the size of it.

.tran 5n 40u uic

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